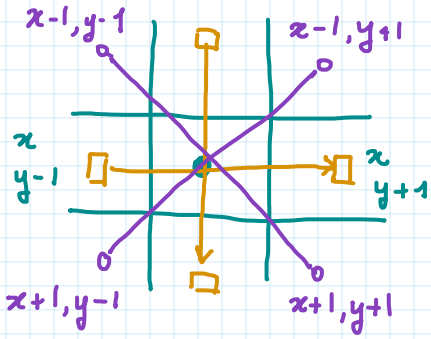

FILTRO di SOBEL

$$M \in \mathbb{R}^{3\times 3}$$

$$M_x \qquad M_y$$

$h = k = -1$

$$f(x-1, y-1) \cong f(x,y) - \frac{\partial f}{\partial x}(x,y) - \frac{\partial f}{\partial y}(x,y)$$

$h = k = 1$

$$f(x+1, y+1) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y)$$

① $$\frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y) \cong \frac{f(x+1,y+1) - f(x-1,y-1)}{2}$$

HW : fare le altre APPROX.

$$f(x+h, y+k) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot h + \frac{\partial f}{\partial y}(x,y) \cdot k$$

$h = -1$, $k = +1$.

$$f(x-1, y+1) \cong f(x,y) - \frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y)$$

$h = +1$, $k = -1$.

$$f(x+1, y-1) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) - \frac{\partial f}{\partial y}(x,y)$$

② $$\frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y) \cong \frac{f(x+1,y-1) - f(x-1,y+1)}{2}$$

COMPONENTI

$$\cancel{2}\,\frac{\partial f}{\partial x} \overset{①+②}{=} \frac{\overbrace{f(x+1,y+1)} + \overbrace{f(x+1,y-1)} - \overbrace{f(x-1,y-1)} - \overbrace{f(x-1,y+1)}}{\cancel{2}\,4}$$

HW . $\frac{\partial f}{\partial y}$ ①-②

Direzione ORIZZONTALE:

$h=0$

$$\left.\begin{array}{ll} k=-1 & f(x,y-1) \cong f(x,y) - \frac{\partial f}{\partial y}(x,y) \\ k=1 & f(x,y+1) \cong f(x,y) + \frac{\partial f}{\partial y}(x,y) \end{array}\right\} \Rightarrow \frac{\partial f}{\partial y}(x,y) \cong \frac{f(x,y+1)-f(x,y-1)}{2}$$

Direzione VERTICALE: $k=0$

$$\left.\begin{array}{ll} h=-1 & f(x-1,y) \cong f(x,y) - \frac{\partial f}{\partial x}(x,y) \\ h=1 & f(x+1,y) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \end{array}\right\} \Rightarrow \frac{\partial f}{\partial x}(x,y) \cong \frac{f(x+1,y)-f(x-1,y)}{2}$$

Il filtro è una MASCHERA $M \in \mathbb{R}^{3\times 3}$

$$M_{ij} = |M^{(x)}_{ij}| + |M^{(y)}_{ij}|$$

$M^{(x)}$

DIAGONALE	VERTICALE	ORIZZONTALE
$+f(x+1,y-1)$ /4	$+f(x+1,y)$ /2	0
$-f(x-1,y+1)$ /4	$-f(x-1,y)$ /2	
$+f(x+1,y+1)$ /4	2 VOLTE	
$-f(x-1,y-1)$ /4		

$$M_x = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

HW.

$$M_y = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

NOTAZIONE: L immagine LUMINANZA

L_x risultato della derivata rispetto a x. $L * M_x = L_x$

L_y risultato della derivata rispetto a y $L * M_y = L_y$

$$\begin{array}{l} x-1,y-1 \cdot m_{11} + x-1,y \cdot m_{12} + x-1,y+1 \cdot m_{13} + \\ x,y-1 \cdot m_{21} + x,y \cdot m_{22} + x,y+1 \cdot m_{23} + \\ x+1,y-1 \cdot m_{31} + x+1,y \cdot m_{32} + x+1,y+1 \cdot m_{33} \end{array} = \text{Nuovo VALORE da mettere in } x,y$$